Given that $y = x + 3$. (Denominator = Numerator + 3).

Therefore, the fraction can be written as $\frac{x}{x+3}$. As per the given condition, $\frac{x+2}{(x+3)+9} = \frac{5}{6}$

By cross multiplication, $6(x+2) = 5(x+3+9)$

$$6x+12 = 5(x+12)$$
$$6x+12 = 5x+60$$
$$6x - 5x = 60 - 12$$
$$x = 60 - 12$$
$$x = 48.$$

Therefore, the original fraction is $\frac{x}{x+3} = \frac{48}{48+3} = \frac{48}{51}$.

Example 3.39

The sum of the digits of a two-digit number is 8. If 18 is added to the value of the number, its digits get reversed. Find the number.

Solution:

Let the two digit number be xy (i.e., ten's digit is x, ones digit is y)

Its value can be expressed as $10x+y$.

Given, $x+y = 8$ which gives $y = 8 - x$

Therefore its value is $10x+y$

$$= 10x + 8 - x$$
$$= 9x + 8.$$

The new number is yx with value is $10y + x$

$$= 10(8 - x) + x$$
$$= 80 - 9x$$

Given, when 18 is added to the given number (xy) gives new number (yx)

$$(9x + 8) + 18 = 80 - 9x$$
$$\text{This simplifies to } 9x + 9x = 80 - 8 - 18$$
$$18x = 54$$
$$x = 3 \Rightarrow y = 8 - 3 = 5$$
$$\text{The two digit number is } xy = 10x+y \Rightarrow 10(3)+5 = 30+5 = 35$$

Example 3.40

From home, Rajan rides on his motorbike at 35 km/hr and reaches his office 5 minutes late. If he had ridden at 50 km/hr, he would have reached his office 4 minutes earlier. How far is his office from his home?

Solution:

Let the distance be 'x' km. (Recall that, time = $\frac{\text{Distance}}{\text{Speed}}$)

Speed 1 = 35 km/hr
Speed 2 = 50 km/hr

Time taken to cover 'x' km at 35 km/hr: $T_1 = \frac{x}{35}$ hr

Time taken to cover 'x' km at 50 km/hr: $T_2 = \frac{x}{50}$ hr

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